ABHAY DESHMUKH

ECE STUDENT | ENTRY-LEVEL PCB DESIGN & HARDWARE ENGINEERING

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PROFESSIONAL SUMMARY

Final-year Electronics & Communication Engineering student with hands-on experience in embedded systems, microcontroller programming, sensor interfacing, and hardware-oriented monitoring projects. Built an ESP32-based multi-sensor node and integrated sensor data with cloud services over MQTT. Interested in entry-level PCB design and electronics hardware roles, with a foundation in electronics, system integration, testing, and technical documentation. Eager to develop practical skills in schematic capture, PCB layout, design-rule checks, and board validation.

TECHNICAL SKILLS

Embedded & Hardware: ESP32, Arduino, Embedded C, sensor interfacing, microcontroller peripherals, hardware setup and integration

Sensors & Instrumentation: ADXL345 vibration sensor, MPX5700AP pressure sensor, DHT22 temperature/humidity sensor, MAX4466 noise sensor, IR and proximity sensors

Connectivity & Data: MQTT, AWS IoT Core, OPC UA, Modbus (RTU/TCP), industrial networking

Automation & Control: Allen-Bradley MicroLogix 1100, ladder logic, sensor-based interlocks, control sequencing

Engineering Practices: Hardware integration, system testing, fault detection, technical documentation, predictive maintenance

PCB Design Tools: Add the tools you have actually used (e.g., KiCad, Altium Designer, or OrCAD) before applying; these were not listed in the supplied resume.

RELEVANT PROJECTS

IoT-Enabled Predictive Maintenance System

- Built an ESP32-based monitoring node integrating five sensors for temperature, vibration, pressure, humidity, and noise measurements.
- Connected sensor data to AWS IoT Core using MQTT and prepared a hardware setup and deployment guide.
- Evaluated ANN and ensemble machine-learning models for fault detection; the ANN model achieved the best classification accuracy.

Intelligent PLC-Based Industry 4.0 Smart Manufacturing System

- Developed an industrial monitoring concept combining PLC control logic with OPC UA and MQTT for machine-data exchange.
- Added an ANN-based fault-classification layer and documented the system in an IEEE-format technical paper.

Automatic Car Parking System

- Programmed an Allen-Bradley MicroLogix 1100 PLC to control slot detection, gate operation, and vehicle counting.
- Integrated IR and proximity sensors for real-time occupancy detection and automated gate sequencing.

EXPERIENCE

Embedded Systems Intern — Pantech Prolabs India | 20 February – 20 May

- Completed hands-on training in Embedded C, microcontroller peripheral interfacing, and firmware development.

EDUCATION

B.Tech, Electronics & Communication Engineering | Expected April 2027

Tulsiramji Gaikwad-Patil College of Engineering and Technology (TGPCET), Nagpur | CGPA: 8.7

HSC (12th): Jawahar Navodaya Vidyalaya, Yavatmal — 83.60% **SSC (10th):** Jawahar Navodaya Vidyalaya, Yavatmal — 89.60%